

Reads: (R^m, x, v) , where $m \in \{\text{na}, \text{rlx}, \text{acq}, \text{sc}\}$

Writes: (W^m, x, v) , where $m \in \{\text{na}, \text{rlx}, \text{rel}, \text{sc}\}$

RMWs: (U^m, x, v_o, v_n) or $(U^m, x, v_o, \perp)$, where $m \in \{\text{rlx}, \text{acq}, \text{rel}, \text{acq-rel}, \text{sc}\}$

Fences: F^m , where $m \in \{\text{acq}, \text{rel}, \text{acq-rel}, \text{sc}\}$

Given an event e , we write $\text{mod}(e)$ for the mode of e .

“Mode strength” order $\sqsubseteq$ is given by:



Given a set of events E , and a mode m , we write:

$$E \sqsupseteq^m \triangleq \{e \in E \mid m = \text{mod}(e) \vee m \sqsubseteq \text{mod}(e)\}$$